

10PEARLS Internship

Cohort 9

Student Name: Asad Bashir

Instructor: Miss Hafsa Imtiza, Miss Umema Ashar

Domain: Data Sciences

Duration: 7 Week

Project: 10 PEARLS Karachi AQI Predictor App

Karachi AQI Forecasting Internship Project

1. Project Overview

Karachi AQI Forecasting System is an AI-powered system that forecasts Karachi's **Air Quality Index (AQI) for the next 3 days, hour by hour.**

The system uses historical air-quality and weather data to train Machine Learning models. It then uses data to predict AQI for the next **72 hours**.

Main Goal

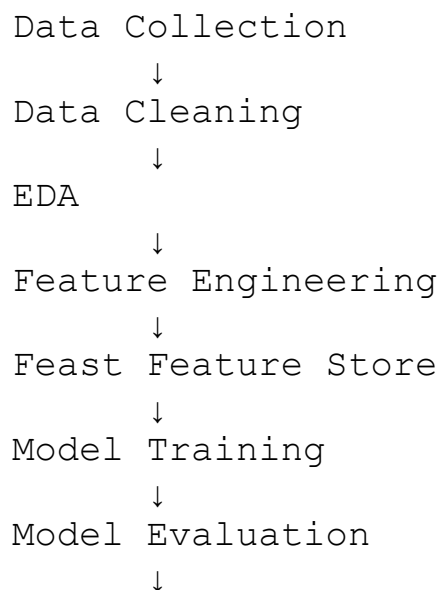
The main goal of this project is to provide an easy way to understand the expected air quality in Karachi for the next three days.

Key Features

- 72-hour AQI forecasting
 - Hour-by-hour predictions
 - Historical AQI analysis
 - Weather and air-quality data collection
 - Feature engineering
 - Multiple Machine Learning models
 - Feast Feature Store
 - Automated CI/CD pipelines
 - Interactive Streamlit dashboard
 - Model training and feature pipelines
-

2. Project Workflow

The complete system follows this workflow:



Best Model Selection



CI/CD Pipeline



Streamlit Dashboard



72-Hour AQI Forecast

3. Data Collection

The project collects historical air-quality and weather data for Karachi from [Open-Meteo.com](https://open-meteo.com).

The data includes information such as:

Air Quality Data

- AQI
- PM2.5
- PM10
- CO
- NO2
- SO2
- O3

Weather Data

- Temperature
- Relative humidity
- Atmospheric pressure
- Wind-related measurements
- Other available weather variables

The historical dataset covers data from **September 2022 to July 2026**.

For the live forecasting system, the latest weather and air-quality data is collected through the **Open-Meteo API**.

4. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset before training the models.

The analysis included:

- Checking missing values
- Checking duplicate records
- Understanding data types
- Analyzing AQI distribution
- Studying AQI over time
- Finding relationships between AQI and weather variables
- Checking correlations between features
- Identifying unusual values and patterns
- Understanding hourly and daily AQI changes

EDA helped in understanding which variables could be useful for forecasting AQI.

5. Data Preprocessing

Before model training, the data was prepared by:

- Sorting data by time
- Handling missing values
- Removing unnecessary data
- Converting timestamps correctly
- Checking data consistency
- Preparing the data for time-series forecasting

The data was kept in time order because this is a forecasting problem.

6. Feature Engineering

Feature engineering was an important part of the project.

New features were created from historical AQI and weather data to help the models understand recent patterns.

Main Features

Lag Features

Previous AQI values were used, such as:

- Previous hour AQI
- Previous several hours AQI
- Previous day AQI

Rolling Features

Rolling statistics were created to understand recent AQI behavior:

- Rolling mean
- Rolling standard deviation

Weather Change Features

Changes in weather variables were also calculated, such as:

- Temperature change
- Humidity change
- Pressure change

These features help the model understand how recent conditions can affect future AQI.

7. Forecasting Target

The system is designed to forecast AQI for the next **72 hours**.

Day 1 → 24 hourly AQI predictions

Day 2 → 24 hourly AQI predictions

Day 3 → 24 hourly AQI predictions

Total → 72 hourly predictions

The model predicts future AQI based on the latest available features and historical patterns.

8. Machine Learning Models

Multiple Machine Learning models were tested and compared.

The project included:

- Ridge Regression
- Naive Persistence
- Seasonal Persistence
- Random Forest
- Extra Trees
- Gradient Boosting
- HistGradientBoosting
- LightGBM
- XGBoost
- CatBoost
- LSTM
- Weighted Ensemble

Different models were evaluated using forecasting metrics such as:

- MAE
- RMSE
- R^2 Score

9. Model Selection

Based on the evaluation of all trained models, a **weighted ensemble blending approach** was selected as the final forecasting model. The ensemble combines the predictions of the **Tuned LightGBM, Tuned XGBoost, and Extra Trees** models.

10. Why Feast Feature Store?

Feast was used as the Feature Store for the project.

The main reason for using Feast was to separate **feature generation** from **model prediction**.

It provides a proper way to:

- Store features
- Manage feature definitions
- Retrieve features for prediction
- Keep training and serving features consistent
- Support an ML production workflow

The project uses Feast for retrieving the latest features before making the 72-hour forecast.

Feast Setup

The project uses:

Offline Store → DuckDB
Online Store → SQLite
Registry → Local Registry

DuckDB was used for the offline feature data because the dataset could be processed efficiently without using a cloud service.

11. Why Feast Instead of Hopsworks?

Hopsworks was initially considered for the Feature Store.

However, the project faced practical issues with the Hopsworks setup and cloud dependency.

For this internship project, **Feast was selected because it could run locally and integrate more easily with the existing project.**

This made the system:

- Easier to develop
- Easier to test
- Less dependent on external cloud services

- Suitable for the internship environment
-

12. Model Training Pipeline

The training pipeline automatically:

1. Loads the prepared data
2. Performs feature engineering
3. Prepares training data
4. Trains the models
5. Evaluates the models
6. Selects/saves the trained models
7. Updates the required project files

The trained models are stored and used by the forecasting dashboard.

13. CI/CD Pipeline

GitHub Actions was used to automate important project tasks.

The project contains automated workflows for:

Feature Pipeline

Runs regularly to:

- Collect/update data
- Generate new features
- Update the Feature Store data

Training Pipeline

Runs regularly to:

- Train the forecasting models
- Evaluate the models
- Update the trained model

CI Validation

The CI workflow checks whether the project setup and required components are working correctly.

CI/CD Flow

```
New Data
  ↓
Feature Pipeline
  ↓
Feature Store
  ↓
Training Pipeline
  ↓
New Model
  ↓
Dashboard
```

This reduces manual work and makes the project closer to a production ML system.

14. Dashboard

A **Streamlit dashboard** was developed to present the forecasting results in an easy-to-understand way.

The dashboard connects the complete ML pipeline:

```
Latest Data
  ↓
Feature Engineering
  ↓
Feast Feature Store
  ↓
Trained Model
  ↓
72-Hour Forecast
  ↓
Dashboard
```

Dashboard Shows

- Current AQI information
- Forecasted AQI
- Hour-by-hour predictions
- 3-day forecast
- Forecast trends
- Latest available information
- Model-generated results

The dashboard provides a simple interface for viewing the predicted air quality without running the ML code manually.

15. Technology Stack

Programming

- Python

Data & ML

- Pandas
- NumPy
- Scikit-learn
- LightGBM
- XGBoost
- CatBoost
- TensorFlow/Keras

Feature Store

- Feast
- DuckDB
- SQLite

Data Source

- Open-Meteo API

Dashboard

- Streamlit

Automation

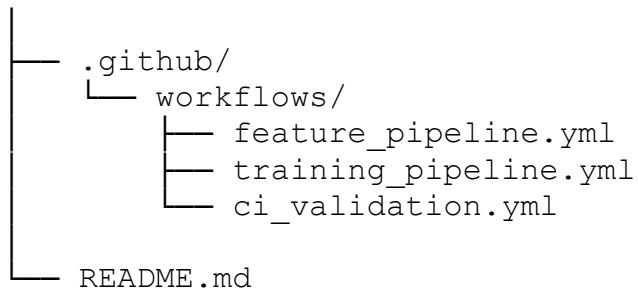
- GitHub Actions
- CI/CD

Version Control

- Git
 - GitHub
 - Git LFS
-

16. Project Structure

```
Karachi-Air-Intelligence/  
├── dashboard/  
│   └── app.py  
├── data/  
│   ├── raw/  
│   └── processed/  
├── feature_store/  
│   ├── data/  
│   └── feature_store.yaml  
├── models/  
│   ├── LightGBM_Tuned.pkl  
│   ├── XGBoost_Tuned.pkl  
│   ├── Extra_Trees.pkl  
│   └── scaler.pkl  
├── notebooks/  
│   └── Karachi_AQI_Forecasting.ipynb  
└── scripts/  
    ├── feature_pipeline.py  
    ├── train_pipeline.py  
    └── verify_setup.py
```



17. Key Project Results

The project successfully provides:

- **3-day AQI forecasting**
- **72 hourly predictions**
- Automated feature generation
- Feature Store integration
- Machine Learning model training
- Model evaluation and selection
- Automated CI/CD pipelines
- Interactive forecasting dashboard

18. Project Link

Streamlit App Link:

<https://karachi-aqi-prediction-app.streamlit.app/>

19. Conclusion

Karachi Air Intelligence is an end-to-end Machine Learning project for **72-hour AQI forecasting in Karachi**.

The project combines:

Data Collection → EDA → Feature Engineering → Feast → Machine Learning → Model Evaluation → CI/CD → Dashboard

The system demonstrates how an ML forecasting model can be developed and connected to an automated pipeline and user-friendly dashboard.